

Test Run Report — Run #148

Environment: Development • Website: aepc2-dev.p2agentx.com • Trigger: Scheduled • Started: Aug 12, 2026 14:00 • Ended: Aug 12, 2026 16:52
Generated: 8/13/2026, 1:54:51 AM

Summary

231

Total Tests

94.2%

Pass Rate (excl. skipped)

178 / 11

Passed / Failed

42

Skipped (connection)

Failures (11)

Failure Cluster	Category	Failures
LLM Reasoning Issues	Llm Reasoning	4
Data Availability Issues	Data Availability	4
Tool Execution Failures	Tool Execution	2
Graph/Visualization Errors	Graph Visualization	1

LLM Reasoning Issues LLM REASONING (4 failures)

Question: When was the last time the BOM data showed up

Error reason: AI indicated failure (matched 'i'm sorry'): I'm sorry, but I can only assist with UI interface actions such as opening views, changing colors, managing settings, or providing guides. For information regarding BOM data history, please contact th

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "When was the last time the BOM data showed up",
    "displayed_response": "I'm sorry, but I can only assist with UI interface actions such as opening view
s, changing colors, managing settings, or providing guides. For information regarding BOM data history, ple
ase contact the plant performance report agent or try a different UI-related request.",
    "coding_response": [
      [
        {
          "answer": "I'm sorry, but I can only assist with UI interface actions such as opening views, chan
ging colors, managing settings, or providing guides. For information regarding BOM data history, please con
tact the plant performance report agent or try a different UI-related request.",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "ui_agent"
    ],
    "used_tools": [],
    "urls": [],
    "steps": []
  }
}
```

Are you satisfied with the answer?

👍 🗨️

Time taken: 6s, token: 0 tokens

When was the last time the BOM data showed up

TE

I'm sorry, but I can only assist with UI interface actions such as opening views, changing colors, managing settings, or providing guides. For information regarding BOM data history, please contact the plant performance report agent or try a different UI-related request.

05:10 PM

Are you satisfied with the answer?

👍 🗨️

Time taken: 11s, token: 0 tokens

connected to Galleria Mall

+ How can we help you today? 🗣️ ➡️

Question: How did my plant perform yesterday?

Error reason: AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 05:35 PM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

TECHNICAL EVIDENCE

Exception	NameError: name 'total_energy_forecast_ag_desc' is not defined
Agent / Tool	data_agent / total_energy_forecast_ag
Endpoint	/testuser1/ai_agent_chat
Location	agent_graph.py:916

API call: GET /testuser2/ai_agent_chat → 400

Raw API response:

```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified
and will investigate.",
  "coding_response": [
    "===== ERROR SUMMARY =====\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGEN
T: data_agent\nTOOL: total_energy_forecast_ag\nERROR TYPE: NameError\nERROR MESSAGE: name 'total_energy_for
ecast_ag_desc' is not defined\nLOCATION: agent_graph.py:916\n=====
=====
  ]
}
```

```
===== ERROR SUMMARY =====
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: total_energy_forecast_ag
ERROR TYPE: NameError
ERROR MESSAGE: name 'total_energy_forecast_ag_desc' is not defined
LOCATION: agent_graph.py:916
=====
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)

it's a fault.

05:35 PM

Are you satisfied with the answer?

👍 🗨️

Time taken: 19s, token: 0 tokens

How did my plant perform yesterday?



⋮

Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.

05:35 PM

Do you want to report this error?

YES NO

Time taken: s, token: tokens

● connected to Abdali Mali Barza

+ How can we help you today? 🗣️ ➡️

LEAVE ANY
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

3. KUPCO_HQ_DAJEEJ

Development • 49.1s

Question: Pull up the latest cleaning log

Error reason: AI indicated failure (matched 'an error'): Sorry, there was an error processing your request.
06:29 PM Do you want to report this error? YES NO Time taken: s, token: tokens connected to Kupco Hq Dajeej LEAVE ANY FEEDBACK Mistakes in P2C

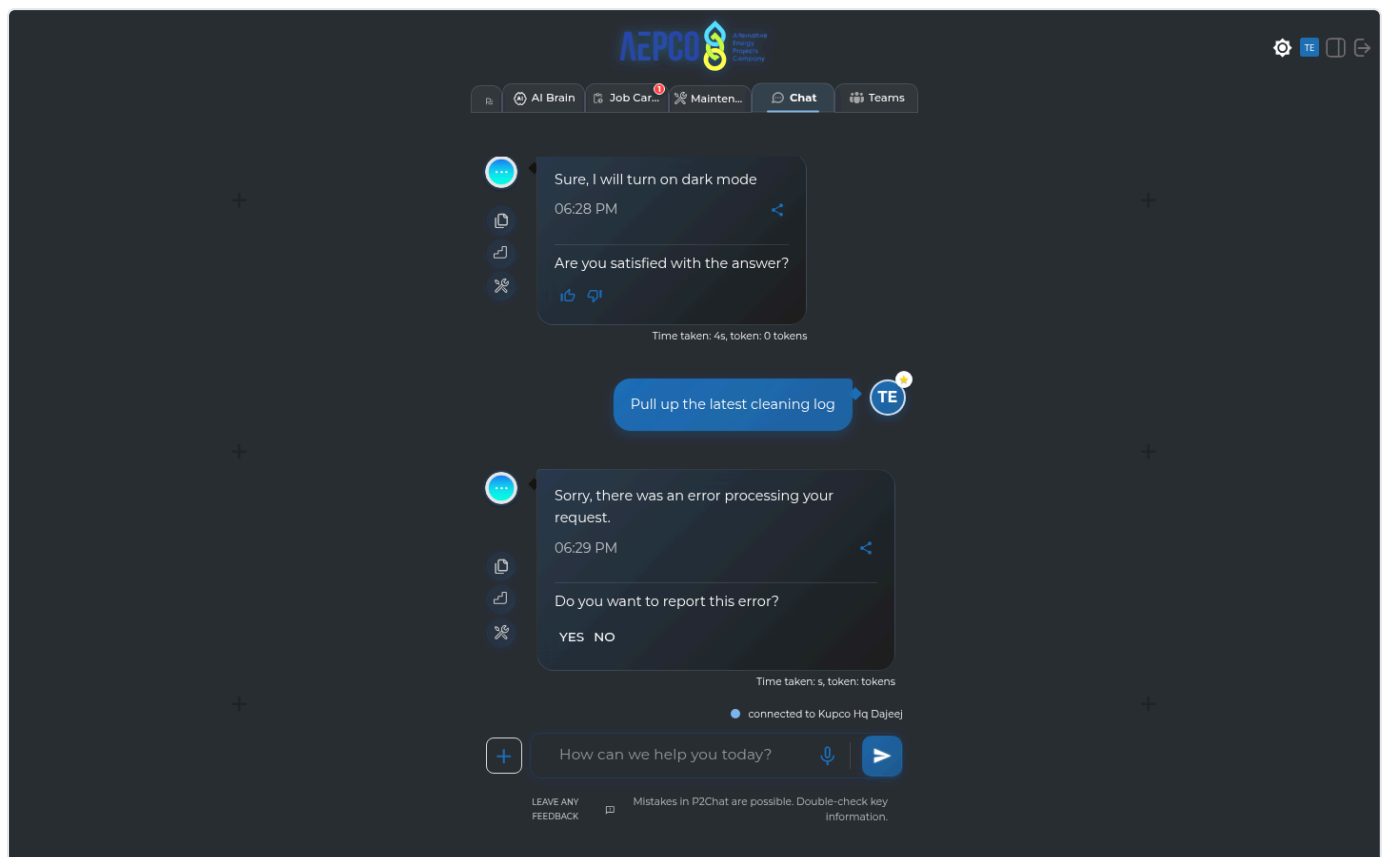
TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 500

Raw API response:

```
{
  "error": "An unexpected error occurred. Please try again later."
}
```

Console errors: Failed to load resource: the server responded with a status of 500 (Internal Server Error)



Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Error reason: AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 07:01 PM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

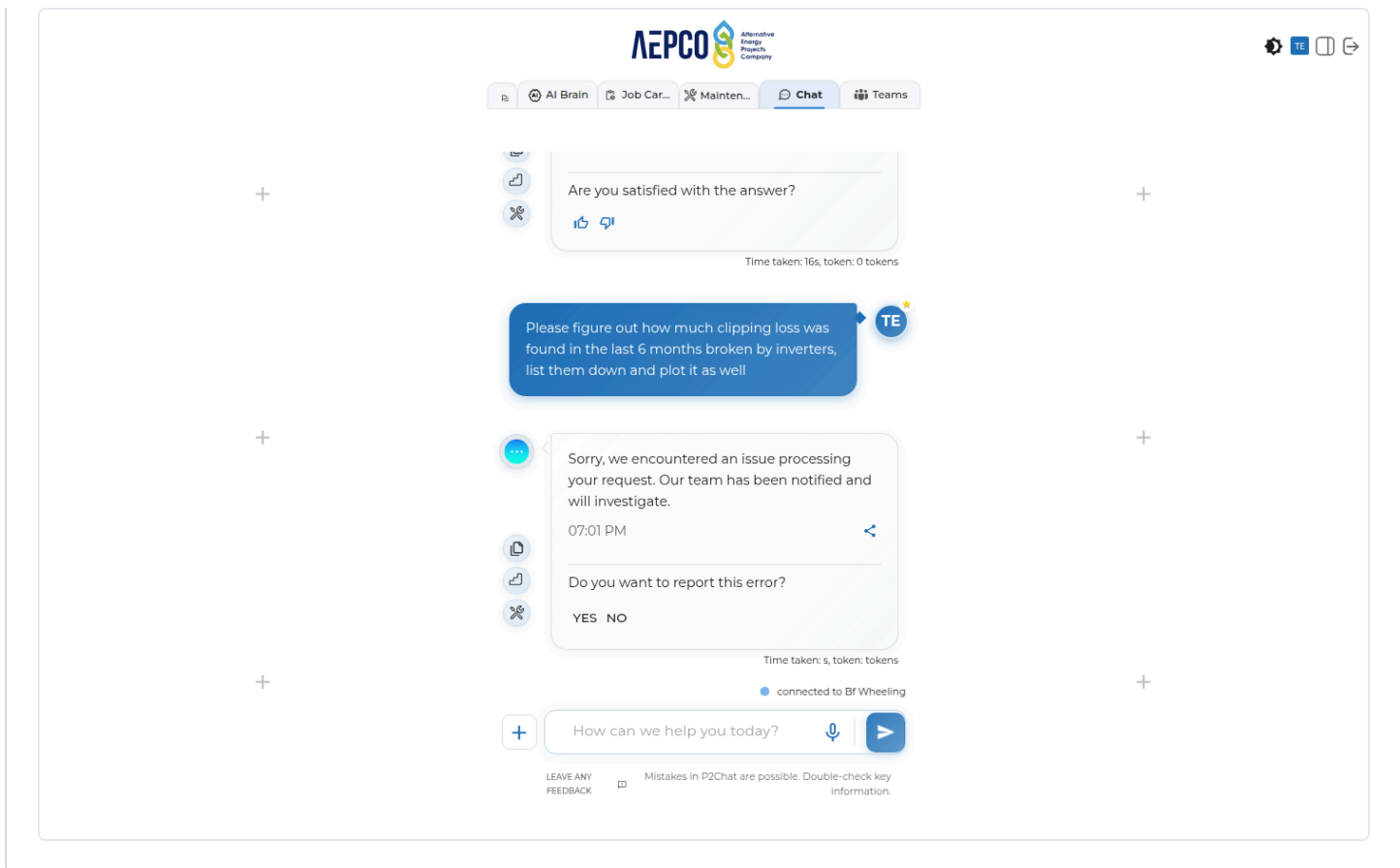
TECHNICAL EVIDENCE

Exception	NameError: name 'total_energy_forecast_ag_desc' is not defined
Agent / Tool	graph_agent / total_energy_forecast_ag
Endpoint	/testuser1/ai_agent_chat
Location	agent_graph.py:606
API call: GET /testuser2/ai_agent_chat → 400	
Raw API response:	

```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.",
  "coding_response": [
    "===== ERROR SUMMARY =====\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGENT: graph_agent\nTOOL: total_energy_forecast_ag\nERROR TYPE: NameError\nERROR MESSAGE: name 'total_energy_forecast_ag_desc' is not defined\nLOCATION: agent_graph.py:606\n=====
  ]
}
```

```
===== ERROR SUMMARY =====
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: graph_agent
TOOL: total_energy_forecast_ag
ERROR TYPE: NameError
ERROR MESSAGE: name 'total_energy_forecast_ag_desc' is not defined
LOCATION: agent_graph.py:606
=====
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)



Data Availability Issues DATA AVAILABILITY (4 failures)

1. KUPCO_HQ_DAJEEJ

Development • 35.3s

Question: How did my plant perform yesterday?

Error reason: AI indicated failure (matched 'no data available'): I can't assess yesterday's performance because there is no data available for 2026-08-11. What I found: - Energy by inverter: inverter1 = NaN, inverter2 = NaN, inverter3 = NaN - Specific yield:

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "I can't assess yesterday's performance because there is no data available for 2026-08-11.\n\nWhat I found:\n- Energy by inverter: inverter1 = NaN, inverter2 = NaN, inverter3 = NaN\n- Specific yield: no data\n- Performance ratio: no data\n- Availability: inverter1 = NaN, inverter2 = NaN, inverter3 = NaN\n- Irradiance (pyranometer_1): no data\n\nSo the plant appears to have no recorded data for yesterday in the database, rather than showing a measurable performance result.",
    "coding_response": [
      "I can't assess yesterday's performance because there is no data available for 2026-08-11.\n\nWhat I found:\n- Energy by inverter: inverter1 = NaN, inverter2 = NaN, inverter3 = NaN\n- Specific yield: no data\n- Performance ratio: no data\n- Availability: inverter1 = NaN, inverter2 = NaN, inverter3 = NaN\n- Irradiance (pyranometer_1): no data\n\nSo the plant appears to have no recorded data for yesterday in the database, rather than showing a measurable performance result."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "functions.perf_ratio_ag",
      "functions.total_energy_func_ag",
      "functions.extract_timeseries_data_ag",
      "functions.technical_contractual_availability_ag",
      "functions.specific_yield_ag",
      "multi_tool_use.parallel"
    ],
    "urls": [],
    "steps": "1. I select data from 2026-08-11 to 2026-08-11, including both start and end dates. \n2. I find all inverter power columns and their DC rated power values using the plant's metadata. \n3. I set calculation parameters, such as the irradiance threshold (usually 50 W/m²), power threshold (usually 0.1 kW), and the data interval (typically 5 minutes). \n4. For each inverter, I filter the data to only include times when irradiance is above the threshold, then count the total number of valid intervals (T_useful) and the number of intervals when the inverter was down (T_down). \n5. I convert these interval counts into hours based on the data interval length. \n6. I calculate the Technical Availability for each inverter using the formula: (T_useful - T_down) / T_useful × 100. \n7. I return a dictionary showing the availability percentage for each inverter.\n\n---\n\n1. I select data from 2026-08-11 to 2026-08-11, including both start and end dates. \n2. I identify the relevant columns for power, irradiance, and temperature. \n3. I clean the data by removing or flagging any invalid values in the power and irradiance columns. \n4. I calculate the actual energy produced by multiplying power by the time interval (0.0833 hours) and adjusting for units if needed. \n5. I calculate the expected energy output without applying temperature corrections. \n6. If the system is bifacial, I apply extra correction factors based on metadata, such as ground reflectivity, area normalization, and shading losses. \n7. I group the actual and expected energy values by day. \n8. I calculate the performance ratio for each day, including bifacial corrections if needed. \n9. I return a detailed breakdown of the performance ratio for each day. \n10. I cap the performance ratio values at 1.3 to avoid unrealistic results.\n\n---\n\n1. I select data from 2026-08-11 to 2026-08-11, including both start and end dates. \n2. I identify all inverter power columns in the dataset, excluding columns for reactive power and flags. \n3. I clean the data by removing or flagging invalid values in each inverter's power column. \n4. For each inverter, I calculate energy by multiplying power by the time interval (0.0833 hours) and adjusting for units (for example, dividing by 1000 if the data is in Watts). \n5. I include all days in the calculation for each inverter. \n6. I sum up all the energy values for the selected period to get a single total energy value for each inverter. \n7. I organize the results into a DataFrame, with inverters as column"
```

- Energy by inverter: inverter1 = NaN, inverter2 = NaN, inverter3 = NaN

- Specific yield: no data

- Performance ratio: no data

- Availability: inverter1 = NaN, inverter2 = NaN, inverter3 = NaN

- Irradiance (pyranometer_1): no data

So the plant appears to have no recorded data for yesterday in the database, rather than showing a measurable performance result.

06:33 PM

Are you satisfied with the answer?

👍 🗣️

Time taken: 21s, token: 0 tokens

● connected to Kupco Hq Dajeej



How can we help you today?



LEAVE ANY
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

2. BF_Wheeling

Development • 55.1s

Question: Please generate last months report

Error reason: AI indicated failure (matched 'error occurred'): Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv.

Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please generate last months report",
    "displayed_response": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AL_Mushagar'",
    "coding_response": [
      [
        {
          "answer": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AL_Mushagar'",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "report_agent"
    ],
    "used_tools": [
      "agent_external_report_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```



Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AL_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AI_Mushagar'

06:54 PM



Are you satisfied with the answer?



< Click here for Source

< Click here for Steps

< Click here for Tools

Time taken: 39s, token: 0 tokens

● connected to BF Wheeling



How can we help you today?



LEAVE ANY
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

3. BF_Wheeling

Development • 53.6s

Question: Please generate December 2025 performance report

Error reason: AI indicated failure (matched 'error occurred'): Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv.

Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please generate December 2025 performance report",
    "displayed_response": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AL_Mushagar'",
    "coding_response": [
      [
        {
          "answer": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AL_Mushagar'",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "report_agent"
    ],
    "used_tools": [
      "agent_external_report_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```

Please generate December 2025 performance report

TE



Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AL_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AI_Mushagar'

07:05 PM



Are you satisfied with the answer?



Time taken: 39s, token: 0 tokens

● connected to BF Wheeling



How can we help you today?



LEAVE ANY
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

4. Abdali_Mall_Netmetering

Development • 87.2s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Error reason: AI indicated failure (matched 'no data available'): Here's the clipping-loss summary broken down by inverter for the last six months (2026-04 through 2026-08-12, note August is partial): • Inverter 1: 94 842.39 kWh • Inverter 2: 5 058.20 kWh •

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well",
    "displayed_response": "Here's the clipping-loss summary broken down by inverter for the last six months (2026-04 through 2026-08-12, note August is partial):\n\n Inverter 1: 94 842.39 kWh\n Inverter 2: 5 058.20 kWh\n Inverter 3: 1 910.82 kWh\n Inverter 4: 3 830.49 kWh\n Inverter 5: 3 990.76 kWh\n Inverter 6: 41 8.45 kWh\n Inverter 7: no data available\n Inverter 8: 2 280.49 kWh\n Inverter 9: 2.93 kWh (only Apr & May)\n Inverter 10: 1 248.22 kWh\n Inverter 11: 0.69 kWh (only Apr & May)\n Inverter 12: 10.29 kWh (only Apr & May)\n Inverter 13: 32.23 kWh (only Apr & May)\n Inverter 14: 2 506.11 kWh\n Inverter 15: 2 007.45 kWh\n\nNote: Data for 2026-08 covers only April 1 through August 12. \n\nI've created a grouped bar chart showing the monthly clipping loss per inverter for the requested period.",
    "coding_response": [
      "Clipping loss by inverter for the last 6 months (monthly, kWh):\n\n2026-04\n inverter1: 12494.63\n inverter2: 554.56\n inverter3: 263.35\n inverter4: 339.32\n inverter5: 407.81\n inverter6: 48.52\n inverter7: NaN\n inverter8: 268.33\n inverter9: 2.41\n inverter10: 131.83\n inverter11: 0.47\n inverter12: 7.25\n inverter13: 14.17\n inverter14: 268.40\n inverter15: 211.93\n\n2026-05\n inverter1: 21526.25\n inverter2: 1069.62\n inverter3: 366.35\n inverter4: 851.81\n inverter5: 888.20\n inverter6: 123.55\n inverter7: NaN\n inverter8: 504.69\n inverter9: 0.52\n inverter10: 322.58\n inverter11: 0.22\n inverter12: 3.04\n inverter13: 18.06\n inverter14: 565.17\n inverter15: 513.16\n\n2026-06\n inverter1: 288 16.52\n inverter2: 1597.10\n inverter3: 393.84\n inverter4: 1270.93\n inverter5: 1319.78\n inverter6: 132.87\n inverter7: NaN\n inverter8: 710.86\n inverter9: NaN\n inverter10: 458.11\n inverter11: NaN\n inverter12: NaN\n inverter13: NaN\n inverter14: 888.63\n inverter15: 753.33\n\n2026-07\n inverter1: 262 72.28\n inverter2: 1475.67\n inverter3: 792.51\n inverter4: 1117.37\n inverter5: 1179.02\n inverter6: 104.57\n inverter7: NaN\n inverter8: 635.94\n inverter9: NaN\n inverter10: 285.16\n inverter11: NaN\n inverter12: NaN\n inverter13: NaN\n inverter14: 681.57\n inverter15: 463.00\n\n2026-08 (partial through 2026-08-12)\n inverter1: 5732.71\n inverter2: 361.25\n inverter3: 94.77\n inverter4: 251.06\n inverter5: 195.95\n inverter6: 8.94\n inverter7: NaN\n inverter8: 160.67\n inverter9: NaN\n inverter10: 50.54\n inverter11: NaN\n inverter12: NaN\n inverter13: NaN\n inverter14: 102.34\n inverter15: 66.03\n\nNote: data available for this calculation starts on 2026-04-01 00:05:00, so February and March are not available.",
      {
        "type": "graph",
        "series": [
          "<html>\n<head><meta charset=\"utf-8\" /></head>\n<body>\n      <div>
        <script type=\"text/javascript\">window.PlotlyConfig = {MathJaxConfig: 'local'};</script>\n      <script charset=\"utf-8\" src=\"https://cdn.plot.ly/plotly-3.0.1.min.js\"></script>
        <div id=\"84afb69a-833b-4804-affd-0ae5b0894c75\" class=\"plotly-graph-div\" style=\"height:100%; width:100%;\"></div>
        <script type=\"text/javascript\">
          window.PLOTLYENV=window.PLOTLYENV || {};\n          if (document.getElementById(\"84afb69a-833b-4804-affd-0ae5b0894c75\")) {\n            Plotly.newPlot(\n              \"84afb69a-833b-4804-affd-0ae5b0894c75\",
              [{\"hovertemplate\": \"%{fullData.name}: %\n            {y:.2f} kWh\\u003cextra\\u003c\\u003c\\u002fextra\\u003e\", \"name\": \"Inverter 2\", \"x\": [\"2026-04-30T00:00:00\", \"2026-05-31T00:00:00\", \"2026-06-30T00:00:00\", \"2026-07-31T00:00:00\", \"2026-08-31T00:00:00\"], \"y\": {\"dtype\": \"f8\", \"bdata\": \"XFEsNH9UGUBIIgOddraQQBrzBgFn9JhAI3uIVqo0l0C2GW8Q9pN2QA=\"}], \"type\": \"bar
          }
        ]
      ]
    ]
  }
}
```

Console errors: An iframe which has both allow-scripts and allow-same-origin for its sandbox attribute can escape its sandboxing.

Question: How did my plant perform yesterday?

Error reason: AI indicated failure (matched 'tool error'): Yesterday (2026-08-11), your plant produced 4,045.86 kWh. Key performance: - Specific yield: 5.43 kWh/kWp - Estimated PR: unavailable for that day because the PR dataset came back empty - Tec

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "Yesterday (2026-08-11), your plant produced 4,045.86 kWh.\n\nKey performance:\n- Specific yield: 5.43 kWh/kWp\n- Estimated PR: unavailable for that day because the PR dataset came back empty\n- Technical/contractual availability: unavailable due to a tool error\n\nContext:\n- Plant DC size: 745.325 kWp\n- The energy result status was Good.",
    "coding_response": [
      "Yesterday (2026-08-11), your plant produced 4,045.86 kWh.\n\nKey performance:\n- Specific yield: 5.43 kWh/kWp\n- Estimated PR: unavailable for that day because the PR dataset came back empty\n- Technical/contractual availability: unavailable due to a tool error\n\nContext:\n- Plant DC size: 745.325 kWp\n- The energy result status was Good."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "multi_tool_use.parallel",
      "functions.query_plant_details_ag"
    ],
    "urls": [],
    "steps": "1. When you ask about a specific plant detail like \"power (kW DC)\", I check the plant's metadata dictionary, focusing on the \"System\" section, and look for the \"power (kW DC)\" field there.\n\n2. I filter the data to include only records from 2026-08-11. Then, I use the \"active_power\" column, clean out any invalid values, and calculate the energy for each data point by multiplying power by 0.0833 hours and converting units if needed. I sum up all these energy values for the day to give you the total energy produced.\n\n3. I select data from 2026-08-11, identify all inverter power columns and their DC ratings from the metadata, and set calculation parameters like irradiance and power thresholds, plus the data interval (usually 5 minutes). For each inverter, I count the intervals where irradiance is above the threshold, tally up useful and downtime intervals, convert these to hours, and calculate technical availability as a percentage for each inverter.\n\n4. I pick data from 2026-08-11, identify the relevant columns for power, irradiance, and temperature, and clean out any invalid measurements. I calculate the actual energy produced by multiplying power by the time interval and adjusting for units, then estimate the expected energy (without temperature correction). If the system is bifacial, I apply extra corrections for factors like ground albedo and shading. I group the data by day, calculate the performance ratio for each interval (with bifacial correction if needed), cap the ratio at 1.3 to avoid unrealistic values, and return the average performance ratio.\n\n5. I select data from 2026-08-11, group it by your chosen time interval (minute, hour, day, month, or year), and include all days in the calculation. I total up the energy produced using the plant's power data and the 0.0833-hour interval, then normalize this by the plant's installed capacity to get the specific yield in kWh per kWp. I provide a detailed breakdown of the specific yield for each time interval."
  }
}
```

Key performance:

- Specific yield: 5.43 kWh/kWp
- Estimated PR: unavailable for that day because the PR dataset came back empty
- Technical/contractual availability: unavailable due to a tool error

Context:

- Plant DC size: 745.325 kWp
- The energy result status was Good.

07:25 PM



Are you satisfied with the answer?



Time taken: 19s, token: 0 tokens

● connected to Abdali Mall Netmetering



How can we help you today?



LEAVE ANY
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

Question: How did my plant perform yesterday?

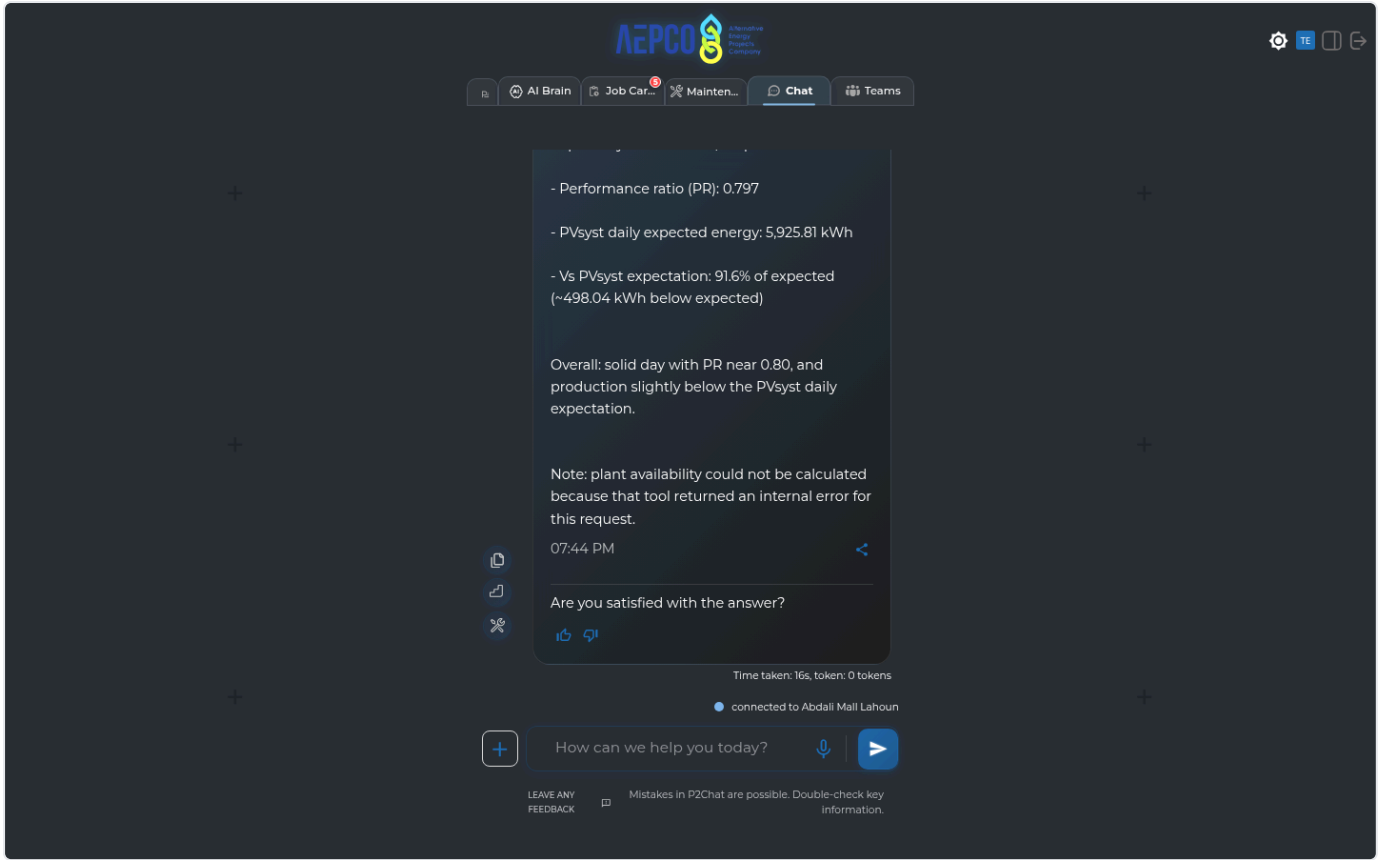
Error reason: AI indicated failure (matched 'internal error'): Yesterday (2026-08-11), your plant performed as follows: - Energy produced: 5,427.77 kWh - Specific yield: 6.91 kWh/kWp - Performance ratio (PR): 0.797 - PVsyst daily expected energy: 5,925.8

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "Yesterday (2026-08-11), your plant performed as follows:\n\n- Energy produced: 5,427.77 kWh\n- Specific yield: 6.91 kWh/kWp\n- Performance ratio (PR): 0.797\n- PVsyst daily expected energy: 5,925.81 kWh\n- Vs PVsyst expectation: 91.6% of expected (~498.04 kWh below expected)\n\nOverall: solid day with PR near 0.80, and production slightly below the PVsyst daily expectation.\n\nNote: plant availability could not be calculated because that tool returned an internal error for this request.",
    "coding_response": [
      "Yesterday (2026-08-11), your plant performed as follows:\n\n- Energy produced: 5,427.77 kWh\n- Specific yield: 6.91 kWh/kWp\n- Performance ratio (PR): 0.797\n- PVsyst daily expected energy: 5,925.81 kWh\n- Vs PVsyst expectation: 91.6% of expected (~498.04 kWh below expected)\n\nOverall: solid day with PR near 0.80, and production slightly below the PVsyst daily expectation.\n\nNote: plant availability could not be calculated because that tool returned an internal error for this request."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "multi_tool_use.parallel",
      "functions.final_result"
    ],
    "urls": [],
    "steps": "1. I select data from August 11, 2026, including that day.\n2. I group the data by day.\n3. I make sure all days in the range are included.\n4. I calculate the total energy produced during this period using the plant's power data, factoring in the 5-minute (0.0833 hour) intervals.\n5. I divide the total energy by the plant's installed capacity (rated power) to get the specific yield in kWh per kWp.\n6. I provide the specific yield for each day as a detailed breakdown.\n\n1. I select data from August 11, 2026, including that day.\n2. I find all inverter power columns and their DC rated powers from the plant's metadata.\n3. I set calculation parameters: irradiance threshold (usually 50 W/m²), power threshold (usually 0.1 kW), and data interval (typically 5 minutes).\n4. For each inverter, I filter the data to times when irradiance is above the threshold, then count the total useful time (T_useful) and downtime (T_down).\n5. I convert these counts to hours based on the data interval.\n6. I calculate each inverter's Technical Availability as (T_useful - T_down) / T_useful * 100.\n7. I return a dictionary showing the availability percentage for each inverter.\n\n1. I select data from August 11, 2026, including that day.\n2. I use the active_power column(s) for calculations.\n3. I clean the data by removing or flagging invalid power measurements.\n4. For each data point, I calculate energy as (Power * 0.0833 hours), converting units if needed (e.g., from Watts to kW).\n5. I include all days in the calculation.\n6. I sum up all the energy values for the selected period to get the total energy produced.\n\n1. I select data from August 11, 2026, including that day.\n2. I identify the relevant columns for power, irradiance, and temperature.\n3. I clean the data by removing or flagging invalid power and irradiance values.\n4. I calculate the actual energy produced by multiplying power by the time interval (0.0833 hours) and adjusting units as needed.\n5. I calculate the expected energy without temperature correction.\n6. If the system is bifacial, I apply extra corrections based on metadata like ground albedo, area normalization, and shading losses.\n7. I group both actual and expected energy by day.\n8. I calculate the performance ratio for each day, applying bifacial corrections if needed.\n9. I return the average performance ratio across all days.\n10. I cap the performance ratio at 1.3 to avoid unrealistic values."
  }
}
```



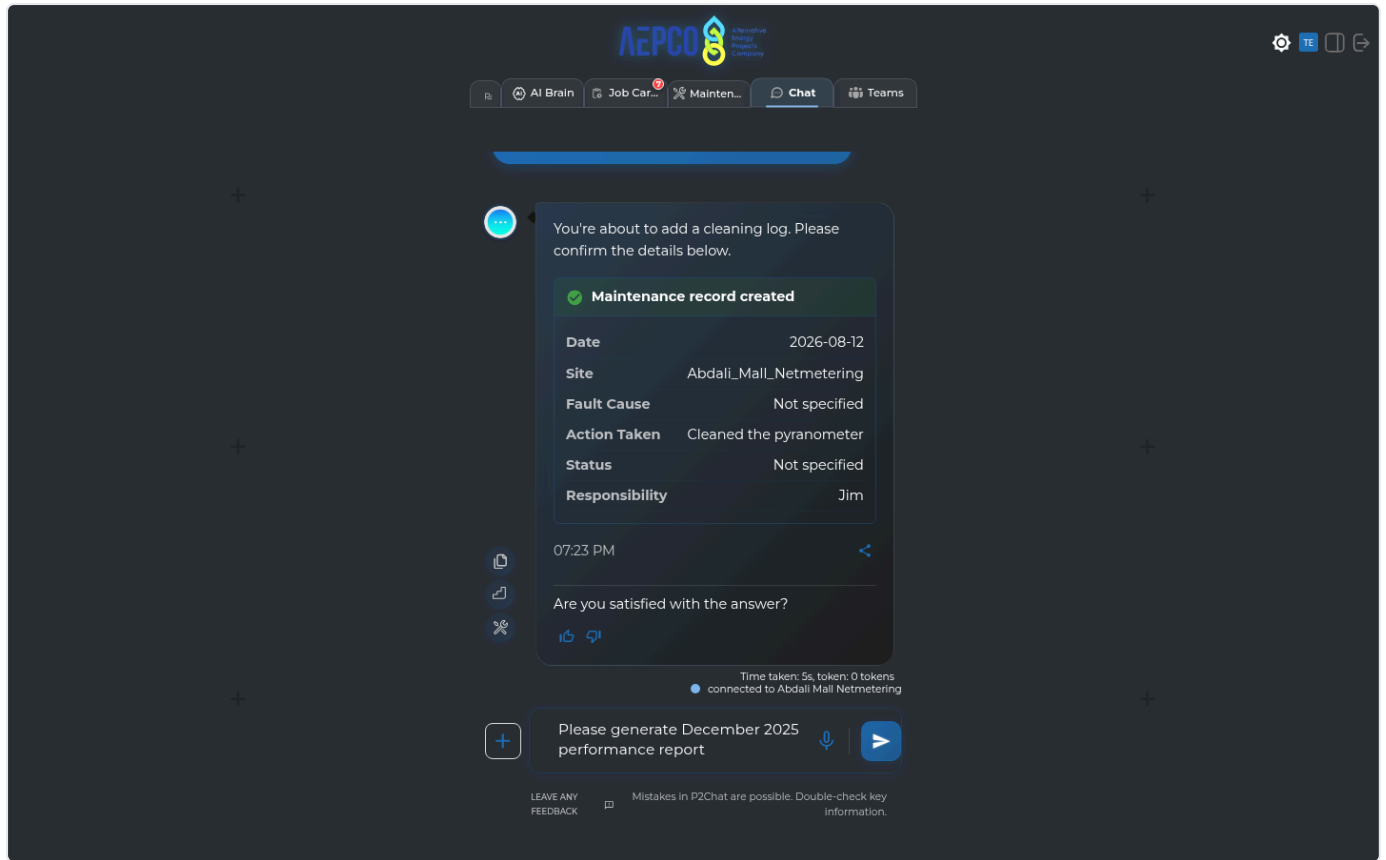
Graph/Visualization Errors GRAPH VISUALIZATION (1 failure)

1. Abdali_Mall_Netmetering

Development • 17.0s

Question: Please generate December 2025 performance report

Error reason: AI indicated failure (matched 'no data available'): AI Brain Job Cards 7 Maintenance Chat Teams Active Power is 14834.58W Last updated: 8/12/2026, 06:55 PM 07:13 PM Good afternoon. How can I help you today? 07:13 PM TE ★ Please plot last 3



Skipped (42)

Plant	Question	Env	Skip reason
SADAFCO	Please generate last months report	Development	Plant connection failed
SADAFCO	Based on pvsyst report, what is the rated power of this plant and array configuration	Development	Plant connection failed
SADAFCO	Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2	Development	Plant connection failed
SADAFCO	Please plot last 3 days hourly performance ratio for my Strings for inverter 2	Development	Plant connection failed
SADAFCO	Please plot the last 4 days hourly performance ratio for all my inverters	Development	Plant connection failed

Plant	Question	Env	Skip reason
SADAFCO	Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3	Development	Plant connection failed
SADAFCO	Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)	Development	Plant connection failed
SADAFCO	How can I shut down my inverters?	Development	Plant connection failed
SADAFCO	I have a COMMUNICATION_FAULT what do I do to my inverters?	Development	Plant connection failed
SADAFCO	When was the last time the BOM data showed up	Development	Plant connection failed
SADAFCO	Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well	Development	Plant connection failed
SADAFCO	Was there any 0 production days in the last 7 days?	Development	Plant connection failed
SADAFCO	Please change to dark mode	Development	Plant connection failed
SADAFCO	Pull up the latest cleaning log	Development	Plant connection failed
SADAFCO	Jim cleaned the pyranometer today, can you add it to the cleaning log?	Development	Plant connection failed
SADAFCO	Please generate December 2025 performance report	Development	Plant connection failed
SADAFCO	How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?	Development	Plant connection failed
SADAFCO	How did my plant perform yesterday?	Development	Plant connection failed
SADAFCO	Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt	Development	Plant connection failed
SADAFCO	Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times	Development	Plant connection failed
SADAFCO	Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well	Development	Plant connection failed
SAK_1	Please generate last months report	Development	Plant connection failed

Plant	Question	Env	Skip reason
SAK_1	Based on pvsyst report, what is the rated power of this plant and array configuration	Development	Plant connection failed
SAK_1	Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2	Development	Plant connection failed
SAK_1	Please plot last 3 days hourly performance ratio for my Strings for inverter 2	Development	Plant connection failed
SAK_1	Please plot the last 4 days hourly performance ratio for all my inverters	Development	Plant connection failed
SAK_1	Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3	Development	Plant connection failed
SAK_1	Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)	Development	Plant connection failed
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